Milestone-4: Incident Type Analysis

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1. Project Description

Milestone-4 focuses on performing a detailed analysis of disaster incident types and integrating insights from previous milestones. The primary objective is to understand how different disaster types are distributed, how they vary across regions, and how they relate to assistance programs.

This milestone represents the final stage of the project, where all analytical components are combined to provide a complete understanding of disaster patterns.

2. Dataset Used

The dataset used for this analysis is the processed FEMA Disaster Declarations dataset, which includes cleaned and enriched features obtained from previous milestones.

Key columns used:

- state
- incidentType
- year
- month
- ihProgramDeclared
- paProgramDeclared
- declarationDate

The dataset contains thousands of records spanning multiple years, making it suitable for comprehensive analysis.

3. Environment Setup

The analysis was performed using Python in a Jupyter Notebook environment.

Libraries used:

- Pandas for data manipulation
- NumPy for numerical operations
- Matplotlib for plotting
- Seaborn for advanced visualization

These tools enabled efficient data processing and visualization.

4. Incident Type Distribution Analysis

The dataset was grouped by *incidentType* to analyze the frequency of different disaster categories.

A bar chart was used to visualize the distribution.

Observations:

- Storms and floods are the most frequent disaster types
- Some disaster types occur significantly less frequently

Interpretation:
This indicates that weather-related disasters dominate the dataset due to environmental and climatic conditions.

5. State Vs Incident Type Analysis

The dataset was grouped by both *state* and *incidentType* to analyze regional variations.

Visualizations used:

- Stacked bar chart
- Heatmap

Observations:

- Different states have different dominant disaster types
- Some states repeatedly experience the same disaster categories

Interpretation:
This reflects the influence of geographical and environmental factors on disaster patterns.

6. Disaster Assistance Program Analysis

The relationship between disaster types and assistance programs was analyzed using:

- ihProgramDeclared
- paProgramDeclared

A grouped bar chart was used for visualization.

Observations:

- Severe disasters trigger assistance programs more frequently
- Public assistance programs are activated more often than individual assistance

Interpretation:
The severity and impact of disasters influence the activation of support programs.

7. Cross Milestone Insight Integration

This stage combines insights from all previous milestones.

- Temporal analysis shows increasing disaster trends
- Geographical analysis highlights regional variations
- Incident analysis identifies dominant disaster types

Result:

A unified understanding of disaster behaviour across time, space, and type.

8. Final Visualization Compilation

The project includes the following visualizations:

Temporal Analysis:

- Yearly disaster trends
- Incident type trends over time

Geographical Analysis:

- State-level disaster maps

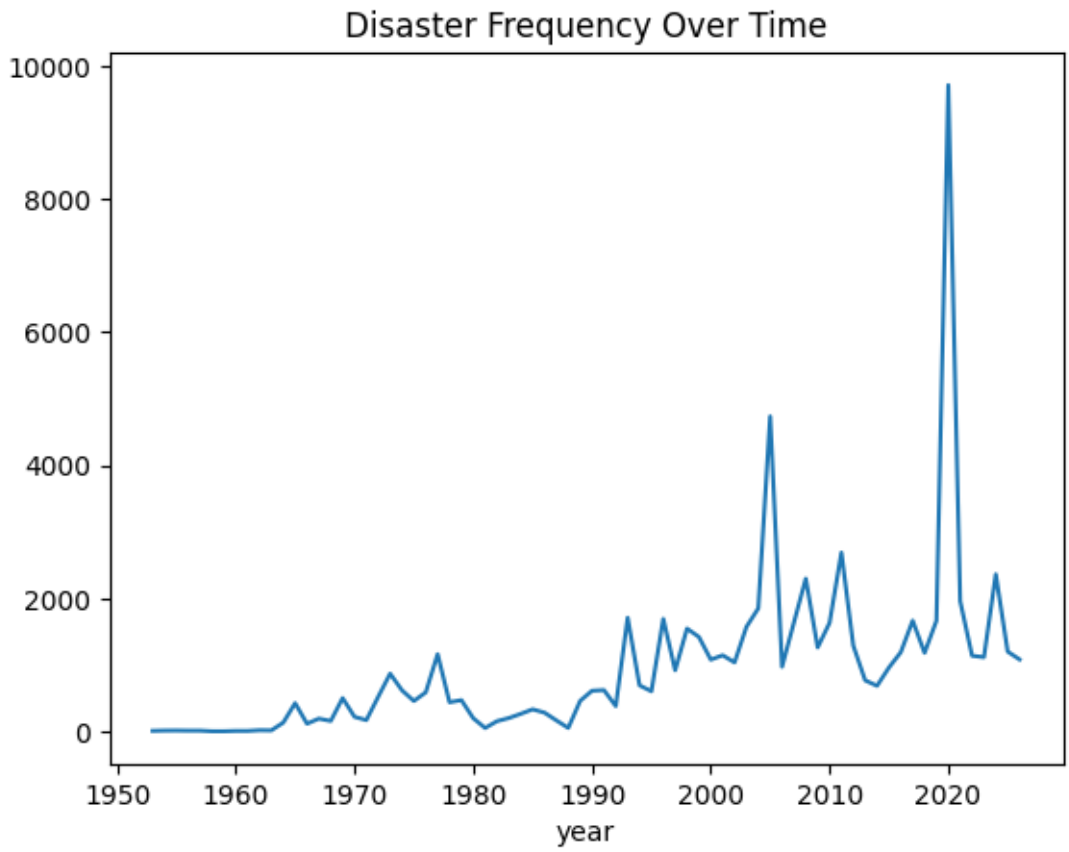
- Disaster hotspot visualization

Incident Analysis:

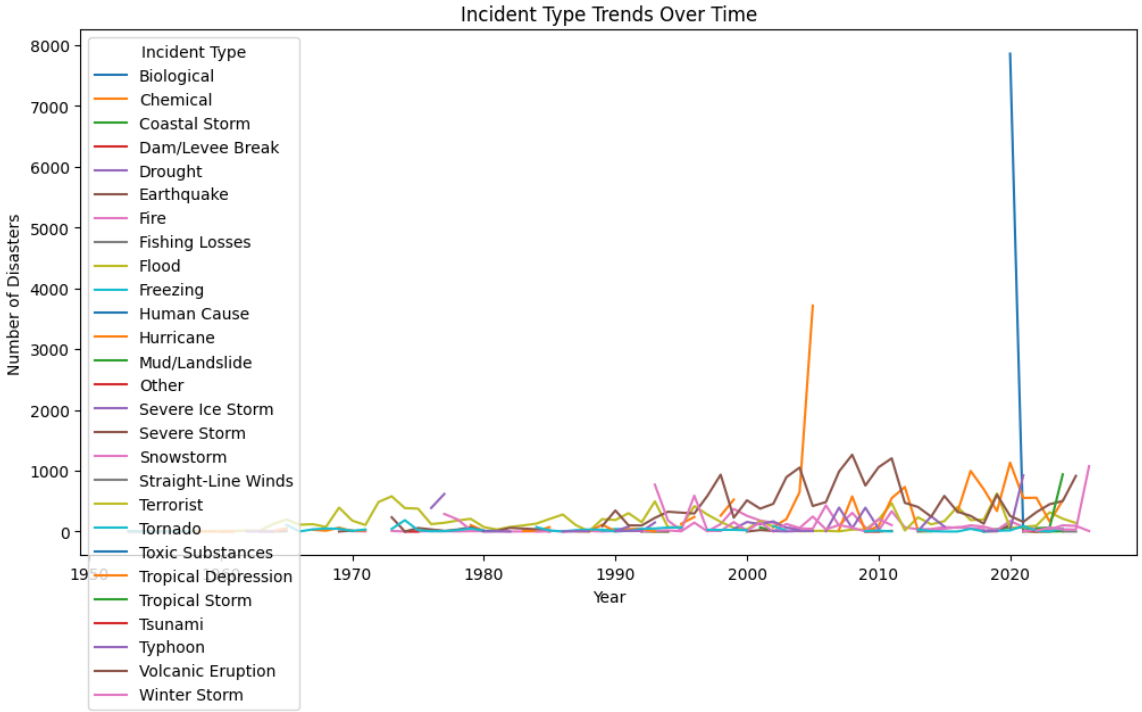
- Incident type distribution
- State vs incident heatmap
- Assistance program comparison

Temporal Analysis:

Yearly disaster trends:

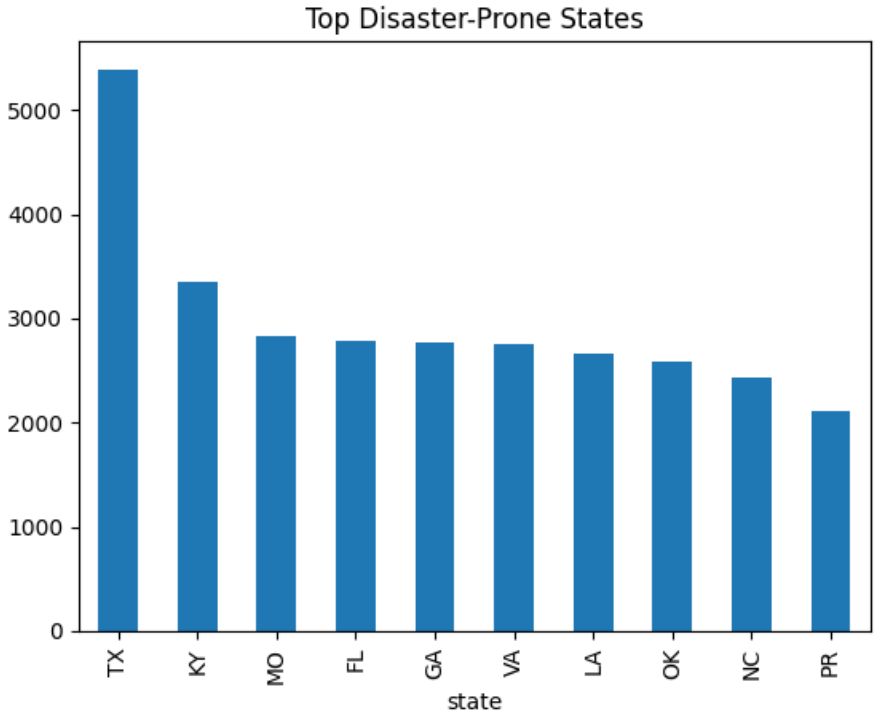


Incident type trends over time

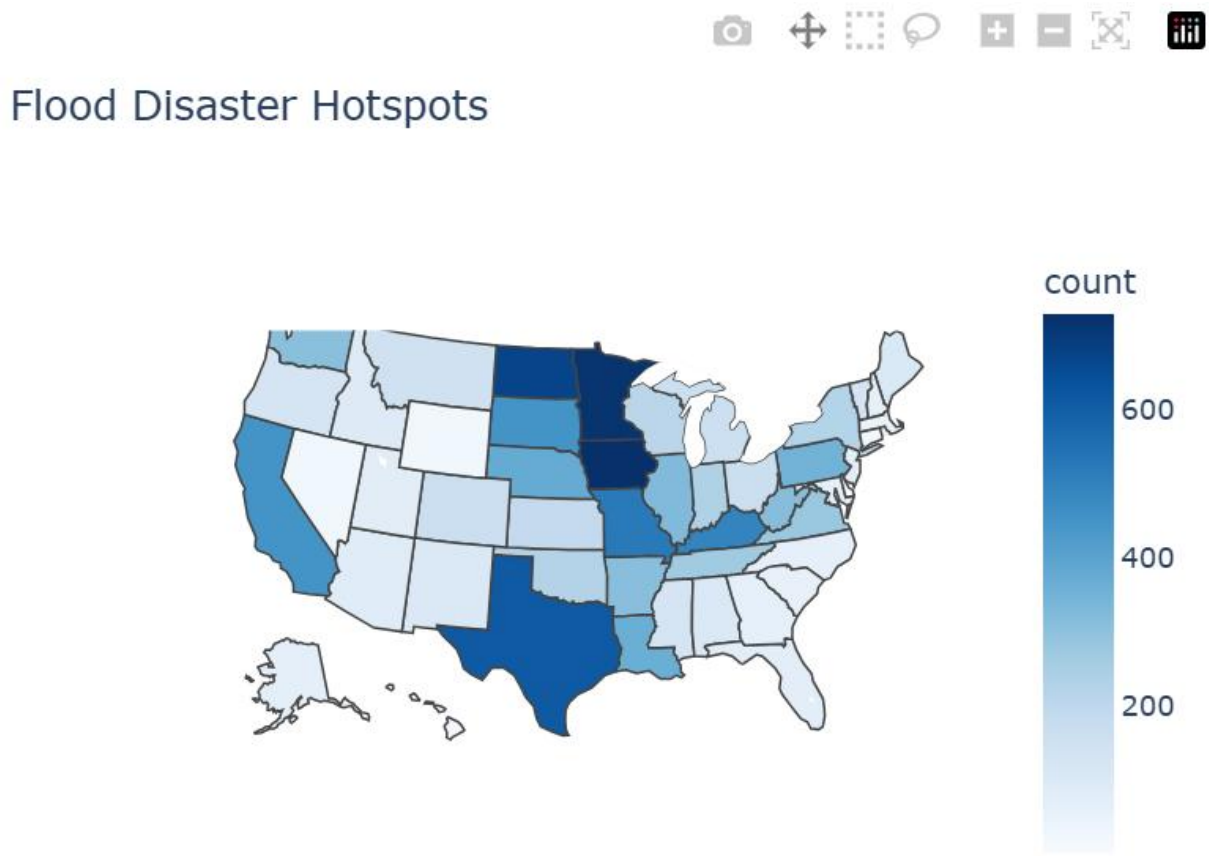


Geographical Analysis:

State-level disaster maps

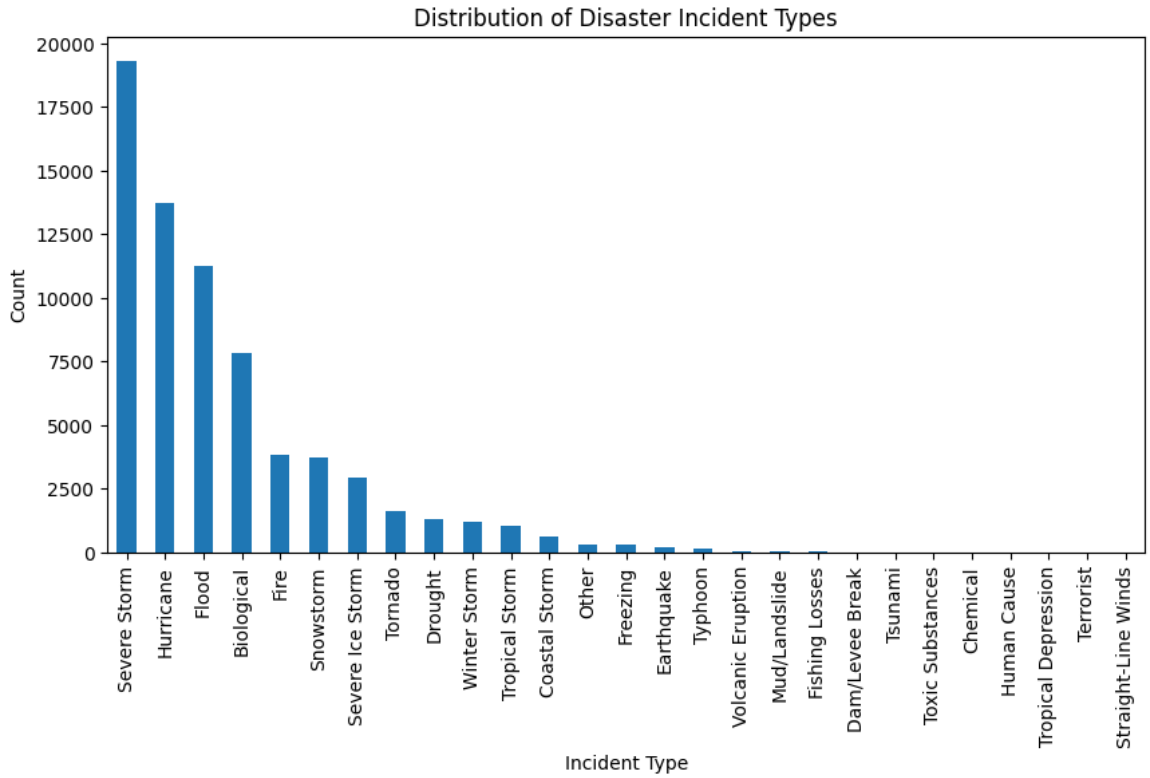


Disaster hotspot visualization

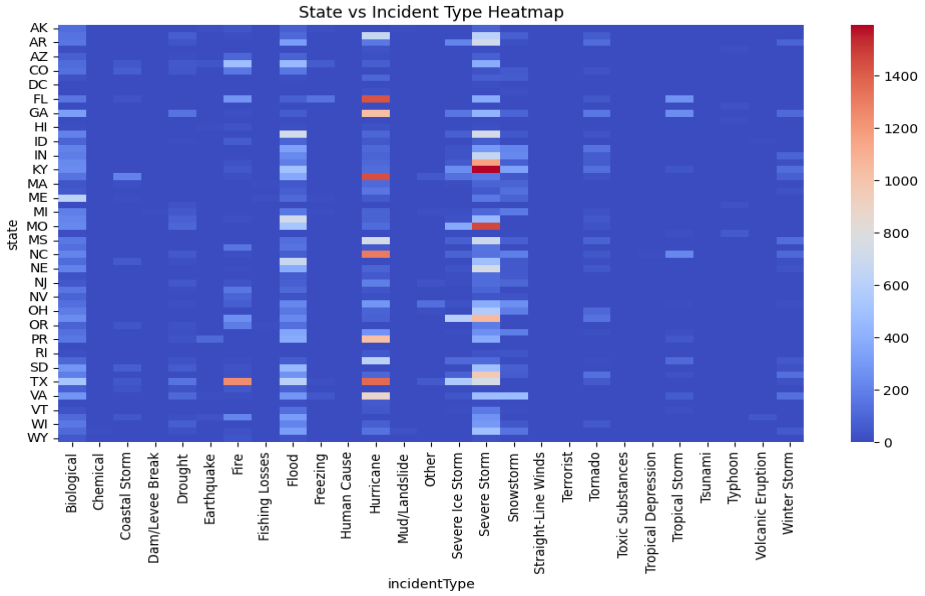


Incident Analysis:

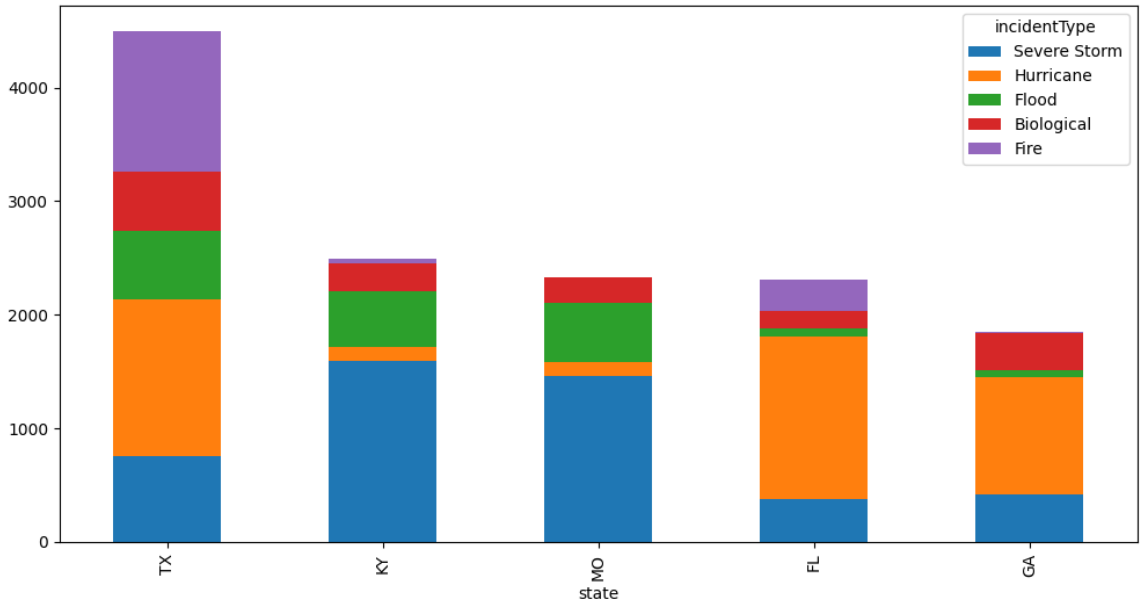
Incident type distribution



State vs incident heatmap



Assistance program comparison



10. Conclusion

The geographical analysis of FEMA disaster declarations provides valuable insights into the spatial distribution of disasters across the United States.

Key findings include:

- Disaster occurrences are unevenly distributed across states
- Coastal and southern states are more disaster-prone
- Certain states consistently experience higher disaster frequencies
- Different regions are affected by different types of disasters
- Environmental and geographical factors significantly influence disaster patterns

This milestone demonstrates the importance of geographical analysis in understanding disaster trends. The insights obtained can support better disaster preparedness, efficient resource allocation, and informed decision-making for disaster management.